Response to the scientific adversarial review — 2026-10-20c

This responds to the internal scientific assessment supplied for this revision; it is not an official WRR editorial decision. The previous manuscript and response are preserved in the 20261020b archive. The new title is **When are incomplete river-temperature records fit for estimating thermal extremes?**

The revision executes the proposed experiments rather than treating additional code or successful builds as scientific answers. It does **not** establish the proposed uncertain-window representation's practical superiority. Unchanged transfer also exceeds its empirical risk target. The manuscript is rewritten around these findings rather than retaining the former method-success narrative.

The protocol, hypothesis table, five-point superiority and two-point equivalence margins, source/target roles, cost grid and stopping decision are in the revision documents. The method lock precedes target calibration errors; the deployment lock precedes new-test errors. A later descriptive snow-unit correction is explicitly recorded and changes no fitted prediction or threshold.

1. Prior work and the actual contribution

We withdraw any implication that previous thermal-data studies considered missing-day counts alone. Johnson et al. (2021) considered timing, and Beaufort et al. (2022) directly reconstructed thermal peaks with seasonal information. Selective prediction with a specified risk target and learn-then-test calibration are also prior statistical ideas. They are credited in the new Introduction and Methods.

The contribution is a fit-for-use assessment connecting paired thermal recovery, an accurately interpreted maximum-statistic identity, matched-information diagnostics, frozen transfer and original-candidate decisions. It is not a new general selective-learning algorithm. A proposed uncertain-window representation is a tested hypothesis, not a demonstrated superior design.

2. Explicit overlap, implicit target information and stronger alternatives

The revision compares H30 disagreement with daily disagreement from the same reconstruction information, and adds seasonal splines, actual reconstruction OOF residual diagnostics, historical mask-matched H30 errors and direct H30 uncertainty. It includes both the current-policy-aligned certificate and a whole-system certificate that selects and calibrates its own delivered point.

On documented-mean source admission, uncertain-window gain over the generic H30 score is 1.7–3.8 percentage points. Its conditional difference from direct H30 uncertainty lies within the predefined two-point band, but whole-procedure uncertainty does not establish equivalence. The complete diagnostic model is competitive too. We do not translate a confidence interval crossing zero into equivalence or interpret a feature deletion as removing implicit peak information.

There is a useful positive result: H30 rather than daily disagreement adds 3.8–6.2 source points and 9.4 points [6.1, 13.1] on independent Swiss records under local recalibration. Both risk point estimates in that latter comparison are below 5%, but their uncertainty does not certify the cap. Explicit overlap improves a restricted logistic model, not consistently a spline. This supports target-specific diagnosis, not the necessity of a particular peak-window representation.

3. Representativeness and the experimental population

The mechanism experiment no longer selects the first twenty-four stations of two providers. It uses all eligible opened records: 406,015 retrospective masks and 290,322 earlier-year masks across 193 labels, 581 provider-station identities and 6,170 distinct station-years. All target and donor identities reduce those labels to 181 overlap components. These components are not asserted to have independent climates.

The primary documented-mean controlled analysis has 312,825 masks on 4,720 station-years in 144 components. Repeated starts, within-warm controls, equal-budget fragmentation and infeasible-cell accounting are explicit. At exact true-peak hiding, the documented-mean length contrast uses 396 station-years in 98 components rather than the earlier eleven station-years.

We also corrected a design-level mistake before final testing: a 30/60/90-day mechanism panel is not an appropriate primary admission population. The new admission assay includes short gaps and empirical shapes: 70,011 retrospective and 46,510 earlier-year cases. Earlier controlled-panel classifier gains are retained as diagnostic history, not the headline. The mixture remains an assay rather than an estimate of natural monitoring outage prevalence.

4. Window switching and the sign of the error

The term previously called a false-peak channel is now the nonnegative window-switch increment. The manuscript gives its exact identity and distinguishes net overestimation from compensation for underestimation. In the primary climatology analysis, 60.4% of positive increments still end below true H30. A coarse aligned/misaligned contrast is not identified with this term.

The claim that H30 is damaged only when its true window is hidden is removed. With fully observed true support, another reconstructed window can still exceed it. Length is not dismissed: extending a fully hidden true peak from a 30- to 90-day outage adds 0.13°C [0.10, 0.17] for climatology and 1.24°C [1.12, 1.35] for interpolation.

5. Oracle diagnosis versus deployable improvement

Residual permutation and mean removal are crossed as a two-by-two intervention. Primary climatology MAEs are 0.450°C originally, 0.398 after permutation, 0.215 after mean removal and 0.071 after both; the interaction is −0.091°C. Their percentages are not additive physical-mechanism contributions.

The separate visible-information correction improves policy MAE by only 0.014°C [0.008, 0.019]. It does not demonstrate the large operational improvement suggested by oracle centering. Conditioning on the true maximum can itself induce negative error; general physical cold bias is not inferred from that selected subset.

6. Risk semantics, zero failures and concentrated errors

The empirical bootstrap selector is explicitly heuristic. Four thousand constructed stress repetitions include nearly perfectly correlated within-group failures, uninformative scores, zero-failure calibration and a rare bad group. In the twelve-group population with true failure probability 6%, the empirical selector violates the 5% target in 45.4% of repetitions. This is not an estimate of river failure frequency.

A separate bounded-group KL test targets the ratio of expected group failure-and-admission to expected group admission. It allows arbitrary within-group dependence but requires independent deployment-representative groups, an independently fitted score and training-fixed candidates. It neither protects every subgroup nor survives arbitrary shift. Even zero failures with full admission need 59 groups for one threshold and 118 for twenty-one. Available calibration cannot certify nonzero admission. Empty admission is not risk zero.

Pooled, equal-component, station-year-weighted and per-component outcomes are retained, including denominators and unavailable estimates. No binomial case interval is claimed for repeated masks in a single dependent river.

7. Frozen transfer and genuinely prospective evaluation

Continuous reference histories were reacquired from original agencies after metadata-only splitting. The 177-catchment extreme resource was not mistaken for complete daily truth. The all-old-target/donor audit and strict completeness leave 53 stations: 21 development, 13 calibration and 19 test, including seven Swiss and twelve French test components. French aggregates remain a distinct sensitivity scope.

F0 transfers both head and source threshold; F1 changes only local calibration; F2 fits on new development records and calibrates locally. Target-provider exclusion and USGS-only intercontinental source restrictions are separate checks. Local historical reconstruction is allowed as stated, not selected by hidden-year error.

Primary Swiss F0 keeps 56.1% of 669 cases but has 8.8% selected failures. F1 keeps 47.5% with 4.1% point risk, and F2 44.4% with 3.0%. The F1 risk interval still exceeds 5%. The calendar-time forward head/calibration

use years through 2018 and every local training input is earlier-year-only for 2019–2022 evaluation. Inputs are current-vintage observations/reanalysis, not archived real-time publication vintages. Its Swiss F0 point risk is 9.8%. Small-group intervals do not prove population risk is above 5%, but do not validate unchanged risk transport either.

8. Independent calendar replay and the no-screen claim

The composition bins were inherited from opened analyses, but rates are fitted on opened-source empirical geometries and frozen before new-station calendar replay. The final paper does not claim that the earlier 67 target station-years were independent of development or that absence of target-calendar labels made that analysis blind.

New eligible natural calendars are French aggregates: 168 masks in ten components, with 21.4% unscreened failure against a composition prediction of 26.6%. Their conditional population differs from the old replay; this does not estimate failure across all monitoring years, many of which have no gaps. No qualifying Swiss documented-mean natural-calendar estimate is available. Most-networks-no-screen advice is withdrawn; a network point forecast cannot certify its risk.

9. Hot/cool refusal and the downstream denominator

We evaluate local training-q90 hot labels, fixed 18/20°C labels, threshold proximity, false-hot/missed-hot outcomes, hot/cool refusal, unknowns, coverage and selected-sample frequency distortion. No unknown record is labelled cool, deleted from the denominator or assumed perfectly repaired.

For primary Swiss F0, hot, cool and overall refusal are 46.4%, 43.4% and 43.9%, so the old no-disproportionate-hot-refusal conclusion is not universal. At declared missed-hot cost five and unknown cost 0.05, loss changes by -0.3368 [-0.4059, -0.2720] per original candidate. Other costs can reverse the benefit. These are cost scenarios, not measured ecological benefits or simultaneous claims across the sensitivity grid.

A full observed-window lower bound can identify a hot label without accepting H30 magnitude. The branch is valid relative to the recorded reference, not a certified sensor-error model. It rescues only two primary retrospective and one prospective labels. It is not presented as a large new design advantage.

10. Missing final-protocol work and evidence closure

A post-test deterministic baseline audit also compares always-unknown/hot/cool answers under the unchanged cost grid. At FN5/A0.05, the primary screen's absolute loss is 0.1191, higher than always unknown at 0.05. A large benefit over reconstructing all does not establish operational desirability; some other declared cells beat both at their point estimates, not as a new confirmatory endpoint. No predictor, threshold or original H3 comparator changes.

The formerly missing work has been executed: broader controls; 324 source score tasks; conditional trajectories and direct-H30 alternatives; actual-reconstructor diagnostics; event/cofailure and natural-calendar assays; continuous new histories, discharge and published-event matching; F0/F1/F2 and provider/intercontinental transfer; Air2stream and CSDI references; repeated learning curves; and 1,000 complete learner/calibration resamples per protocol. Multiple masks and split seeds are never counted as new hydrological replications.

The primary five-point representation hypothesis is not supported. Whole-procedure gain intervals include zero and more target groups alone do not solve training/calibration uncertainty. We stop tuning rather than changing the comparator, margin, role assignment or headline population to obtain a win.

A snow-percentage label error was corrected descriptively after the first frozen report. Neither features nor decisions use it; the immutable locks and original outputs remain unchanged, and corrected hydrological tables supersede the raw labels. Groundwater and regulation classes are unverified, not invented.

The manuscript, SI, five main figures, two supplements and generated tables now describe these final experiments. Source bindings and scientific completion checks are separate from software tests. Author details and

a permanent analysis archive still require verified human input; scientific revision is not a claim of submission readiness or guaranteed WRR acceptance.